

TECHNICAL SCALE PLAN

From embedded proof to field-ready adaptive payload

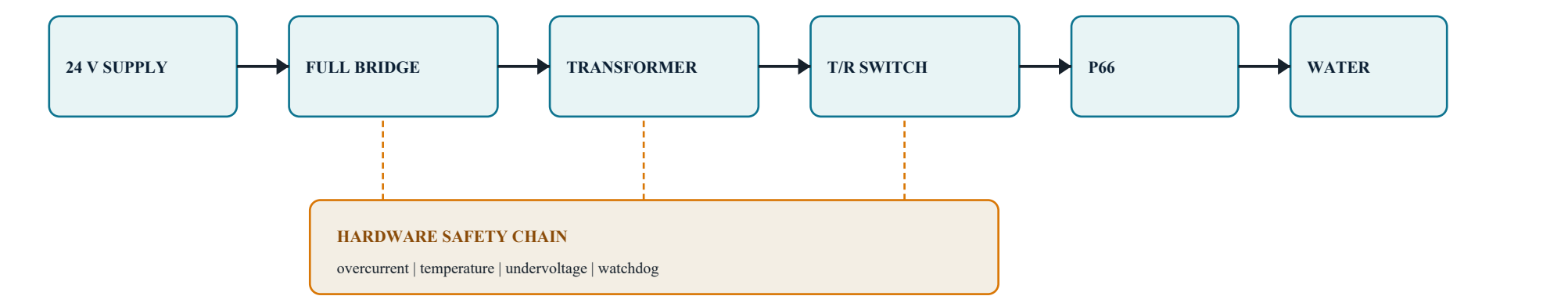
Each stage adds one measurable capability and one procurement commitment. The order protects safety, evidence quality and capital efficiency.



Stage	Build focus	Evidence gate	Commercial claim unlocked
Submission system	ESP32-S3, fixed-point DDS, DMA, guarded solver	Deterministic trace and per-ping energy	Embedded adaptive waveform execution
Integrated tank	Rev-B PCB, sealed rig, T/R protection	Repeatable water-coupled timing and calibration	Paid evaluation and lab pilot
100 W drive	24 V full bridge, transformer, thermal and safety chain	Protected 100 W electrical bursts and recovery	Higher-energy transducer integration
Field payload	Calibrated receive, pressure housing, vehicle interface	Absolute pressure, ingress and operational data	Port, defence and ocean-research pilot
Production	DFM, end-of-line calibration, traceability and service	Yield, acceptance time and release controls	Repeatable procurement and support

POWER AND MEASUREMENT ARCHITECTURE

100 W path, calibrated evidence and budget



Engineering gate	Target	Scaling reserve	INR
Bridge output	100 W burst into characterized load	PCB, drive, transducer and enclosure	2,00,000
Efficiency model	85%; about 18 W burst loss	Receive chain	70,000
Duty-cycle example	10%; about 1.8 W average bridge loss	Calibrated hydrophone package	7,50,000
Transducer reference	Airmar P66, 50/200 kHz, 600 W	Pressure housing and interface	2,50,000
Receive reference	HBK Type 8103, 0.1 Hz-180 kHz	Integration and contingency	1,65,000
Safety	Independent overcurrent, thermal, UV and watchdog	Total	14,35,000

Procurement values are planning allowances. The P66 public price reference is approximately INR 18,600-23,300 before import costs. The hydrophone allowance includes conditioning, calibration documentation and accessories pending a formal quotation.